FAVORS-BERA: A Multi-Modal Biometric Data Architecture for

Resonance-Based Identity Verification in AI-Driven Security Systems

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Abstract

Traditional biometric systems treat identity as a static pattern-matching problem, creating three systemic vulnerabilities: replay attacks against static records, irremediable database compromise, and drift blindness as physiological characteristics evolve over time. This paper introduces FAVORS-BERA, a unified framework reconceptualizing biometric identity as a continuously evolving, multi-dimensional resonance field.

FAVORS (Fingerprint, Aura, Voice, Odor, Retina, Signature) provides six-channel biometric capture with cryptographic forensic timestamping. BERA (Bio-Energetic Resonance Architecture) represents each individual's biometric state as an Aura Resonance Index (ARI) and Emission Resonance Index (ERI), bounded by a personalized Resonance Continuity Index (RCI) replacing population-averaged thresholds, and validated through Resonant Harmony Cycles (RHC). Together these define the Resonance State Record (RSR), enabling trajectory-based identity queries, coherence-space anomaly detection, and privacy-preserving federated verification via the IDIPITIS protocol layer.

Empirical Status: This paper constitutes a complete architectural specification. No prototype or empirical results exist at time of submission. Phase 0 simulation and Phase 1 prototype pathways are fully specified. Collaboration for empirical validation is explicitly invited.

Keywords: *multi-modal biometric identity, resonance-based verification, biometric drift, personalized thresholds, Non-Punitive Remediation*

1. Introduction

The global identity verification challenge is accelerating. As AI systems increasingly gate access to financial infrastructure, healthcare records, national security systems, and civic participation, the adequacy of underlying biometric database architectures has become a foundational security question — not merely a technical one. Three failure modes define the current generation of biometric systems:

- **Static record vulnerability:** a compromised biometric database permanently exposes enrolled individuals, since biometrics cannot be reissued like passwords or cryptographic keys.

- **Replay attack susceptibility:** captured biometric samples can be replayed against static matchers. Six-channel simultaneous multi-modal capture with cryptographic timestamping defeats this class of attack.
- **Temporal drift blindness:** static records cannot detect or accommodate legitimate physiological changes — aging, illness, stress, environmental exposure — that alter an individual's biometric signature over time, producing false rejections and an inability to model identity evolution.

FAVORS-BERA, developed under the New Age Cybernetics (NAC) paradigm at the ERES Institute for New Age Cybernetics (est. 2012), addresses all three failure modes by replacing the static-record model with a dynamic resonance field model. Identity, in this framework, is not a stored artifact but a living coherence state — continuously generated, periodically validated, and inherently multi-modal.

The framework operates under a hard ethical constraint derived from the ERES foundational triad: Don't Hurt Yourself. Don't Hurt Others. Build for Future Generations. This triad is not aspirational language; it is a design specification. Non-Punitive Remediation (NPR) is the architectural response: biometric deviation events are routed to restoration and context clarification, not lockout and punishment. This structural embedding of restorative ethics distinguishes FAVORS-BERA from conventional punitive security paradigms and is reflected directly in the RSR schema's NPR_status field.

2. Background and Related Work

2.1 Limitations of Static Biometric Architectures

Existing multi-modal biometric systems — fingerprint + iris, face + voice — improve accuracy over single-modality systems but do not resolve the static-record problem. Score-level, feature-level, and decision-level fusion approaches all operate on the assumption that each modality produces a fixed enrollment template [1]. The FAVORS-BERA framework departs from this assumption at the architectural level: no template is stored; instead, a continuously updated Resonance State Record tracks the trajectory of each individual's coherence field.

A complete security context requires not only multi-modal capture but a validated methodology for interpreting signal convergence across instruments. The ERES MIEVM (Multi-Instrument Ensemble Validation Method) provides this layer: by routing the same RSR dataset through multiple independent AI analytical engines with complementary bias profiles, MIEVM produces ensemble convergence scores more robust than any single-instrument determination. Applied to FAVORS, MIEVM (in its extended form MIEVM AD_ON-AI / ARI/ERI/RCI) transforms six independent channel readings into a single authoritative identity coherence determination.

2.2 Continuous Authentication and Bio-Signal Dynamics

Research in continuous authentication has explored behavioral biometrics — keystroke dynamics, gait, touch patterns — as temporal signals rather than static templates [2]. More recent work integrates physiological-electric signals: HRV and GSR as continuous authentication proxies have demonstrated strong individual discriminability [Zhao 2025]. Multimodal fusion with personalized thresholds shows measurable improvement over population-averaged approaches [Kumar 2024]. Classification frameworks for continuous biometric authentication [Boshoff 2025] and industry trend analyses [HID Global 2026] confirm industry-wide movement toward dynamic identity models.

FAVORS-BERA extends this trajectory by introducing the resonance field as the fundamental identity data structure — a multi-dimensional biometric state space whose coherence can be

continuously monitored, whose deviations trigger verification events, and whose boundaries are formally defined by the Semiosphere ontological model:

- **Semiosphere** — the Complete Universe+: the totality of all possible biometric signal space, including dimensions not yet measurable.
- **Perciphere** — the boundary of the known Universe: the current frontier of instrumented biometric detection; what FAVORS sensors can reach today.
- **Protosphere** — the ontological core, the working Universe: the actionable identity signal space in which FAVORS-BERA operates — the intersection of what is measurable, interpretable, and verifiable at current instrument capability.

This three-layer model provides a principled basis for anomaly detection calibrated to instrument limits: deviations pushing toward the Perciphere boundary trigger elevated verification rather than immediate rejection.

2.3 New Age Cybernetics Framework

The ERES Institute for New Age Cybernetics has developed a comprehensive framework for bio-energetic measurement and governance, formalized in the cybernetic relationship $C = R \times P / M$ (Cybernetics equals Resources times Purpose divided by Method). BERA, ARI, ERI, RCI, and RHC are components of this framework published across the ERES Proof-of-Work repository and ResearchGate archive [3,4,5,6].

PlayNAC provides the recursive governance kernel that prevents systemic creep — the gradual drift of any resonance-based system toward punitive, centralized, or fear-default structures. Through fractal council structures, merit-resonance weighting, auditable decision records, and automatic ethical-triad veto, PlayNAC ensures that RCI bands cannot be narrowed arbitrarily and that NPR pathways remain structurally open at every access tier.

3. The FAVORS-BERA Architecture

FAVORS-BERA applies the Six Degrees-of-Separation principle structurally: six biometric channels, each providing an independent path to identity resolution, collectively forming a coherence field that no single-channel compromise can defeat.

3.1 FAVORS: Six-Channel Biometric Signal Capture with Forensic Timestamping

FAVORS defines six biometric modalities organized into three signal classes:

- **Physical-structural:** Fingerprint (dermal ridge topology), Retina (vascular branching pattern), Signature (kinematic pen-pressure dynamics and velocity envelope)
- **Bioacoustic-biochemical:** Voice (phonatory resonance spectrum and formant trajectory), Odor (volatile organic compound emission profile)
- **Electromagnetic/Bioelectric:** Aura — in Phase 1, proxied via Heart Rate Variability (HRV), Galvanic Skin Response (GSR), and electrodermal activity [Zhao 2025]; Phase 2 targets full electromagnetic envelope per GAIA EMCI specifications

Six-channel simultaneous capture with cryptographic timestamping provides forensic anchoring functions beyond identity verification:

- **Forensic:** legally admissible, tamper-evident record of presence and biometric condition at a precise moment in time.
- **Emergency Preparedness (EP) and Remediation:** rapid re-identification in mass-casualty or civil emergency scenarios where conventional identification is unavailable.

- **Intellectual Property and Legal:** notarized biometric anchor for authorship, contract execution, and IP origination — the individual's resonance state at moment of creation forms part of the evidentiary record.

3.2 BERA: Bio-Energetic Resonance Architecture with ARI / ERI / RCI

BERA provides the data modeling layer. For each enrolled individual i, BERA maintains:

- **ARI(i,t) — Aura Resonance Index:** normalized vector describing individual i's biometric coherence state at time t. Primary identity signal.
- **ERI(i,t) — Emission Resonance Index:** temporal derivative measuring rate and direction of resonance change between validation periods. Detects drift, stress, illness, or external interference.
- **RCI(i,t) — Resonance Continuity Index:** personalized tolerance band [RCI_min, RCI_max] establishing acceptable variation envelope for ARI and ERI. Personalized, not population-averaged; narrows under high-security contexts, widens under ambient conditions.
- **RHC(i,n) — Resonant Harmony Cycle:** periodic coherence validation event. Current ARI/ERI profile evaluated against individual's historical resonance trajectory and RCI band.

Together ARI, ERI, and RCI form the three-index engine of MIEVM AD_ON-AI: transforming the general multi-instrument ensemble method into a specific biometric coherence engine. The identity verification decision function V(i,t) produces probabilistic confidence intervals, not binary match/no-match, enabling graduated response protocols calibrated to threat context.

3.3 IDIPITIS Integration Layer

IDIPITIS (Internet Protocol Identification Definition Instruction Technology Information Systems) provides the federated protocol layer. Biometric resonance attestations are attached to protocol-level identity tokens as signed, time-bounded credentials, enabling FAVORS-BERA to operate across heterogeneous database environments without centralizing raw biometric data.

Key design principle — pre-authentication relational access control: access to any resource, particularly Real-Time (RT) Media streams, requires FAVORS-BERA identity coherence verification before session establishment, not after. This architectural inversion eliminates the category of attacks that exploit unauthenticated session initialization. The RCI band is carried within the IDIPITIS credential token as a context-sensitive parameter, enabling dynamic access control without round-trips to the central BERA database for each decision.

Table 1 maps all components to DBDM 2026 research tracks:

Table 1. FAVORS-BERA component mapping to DBDM 2026 research tracks.

Component	Function	DBDM 2026 Track
FAVORS	Six-channel biometric capture + forensic timestamp (Fingerprint, Aura/Bioelectric, Voice, Odor, Retina, Signature)	Data Mining Applications: Cybersecurity, Healthcare, Legal/IP
BERA	Bio-Energetic Resonance Architecture — dynamic resonance field modeling	AI for Data Privacy, Security & Trust
ARI	Aura Resonance Index — normalized individual resonance baseline	Feature Engineering & Representation Learning

Component	Function	DBDM 2026 Track
ERI	Emission Resonance Index — temporal resonance drift detection	Anomaly & Outlier Detection
RCI	Resonance Continuity Index — personalized coherence tolerance band (core innovation)	Data Streams & Real-Time Data Management
RHC	Resonant Harmony Cycle — periodic validation event	Data Streams & Real-Time Data Management
IDIPITIS	Federated protocol layer; pre-authentication for RT Media access control	Federated & Heterogeneous Databases
MIEVM AD_ON-AI	Multi-Instrument Ensemble Validation: ARI/ERI/RCI coherence engine; P^3 testing	Data Science Workflows & Human-AI Interaction
NPR	Non-Punitive Remediation — restorative deviation response embedded in RSR	AI Ethics, Fairness, and Responsible Data Use

4. Formal Data Model: The Resonance State Record (RSR)

The RSR is the primary identity data object in FAVORS-BERA. It replaces the static biometric template with a dynamic, trajectory-indexed coherence record. The RSR expresses a State Divide — a binary context of +/− Unity — in which the individual's coherence score is interpreted as either resonant with their baseline (positive unity: continuous access) or deviating from it (negative unity: NPR routing, graduated verification).

RSR(i) = { ARI_vector(t), ERI_delta(t), RCI_band(t), RHC_history[n], FAVORS_stream_refs[], coherence_score(t), state_context(+/−), NPR_status, variable_index{}, threat_context_flag }

Table 2 describes each field:

Table 2. RSR schema — incorporating RCI, NPR_status, and PPDM governance extensions.

RSR Field	Type	Function	Framework
ARI_vector(t)	Float[] + timestamp	Normalized bioelectric resonance baseline at time t	BERA / FAVORS Aura channel
ERI_delta(t)	Float + sign	Rate and direction of resonance change between validation periods	BERA temporal derivative
RCI_band(t)	Range [min, max]	Personalized coherence tolerance — narrows (high-security) or widens (ambient)	RCI — core innovation
RHC_history[n]	Event log[]	Periodic coherence validation record with timestamp and score	RHC Cycle archive
FAVORS_stream_refs[]	URI[] + hash	References to six live biometric channel streams + forensic timestamp anchor	FAVORS six-channel capture

RSR Field	Type	Function	Framework
coherence_score(t)	Float [0.0–1.0]	Computed coherence across all channels weighted by ARI baseline and RCI band	BERA decision layer
state_context (+/−)	Binary enum	Positive: resonant (continuous access). Negative: NPR routing triggered	State Divide / RSR core
NPR_status	Enum + log	Non-Punitive Remediation pathway: none / context-clarify / resource-support / realign	ERES ethical triad
variable_index{}	Map<key,val>	Dynamic registry for governance framework bindings (PPDM / UBIMIA / GCF / NBERS)	PPDM governance layer
threat_context_flag	Enum	Graduated access: continuous / stepped-verify / lockout (temporary, NPR-reviewed)	IDIPITIS access control

The variable_index field enables Privacy-Preserving Data Mining (PPDM) across external governance frameworks — UBIMIA (Universal Basic Income + Merit + Incentives + Awards), GCF (Gracechain Framework), NBERS (National Bio-Energetic Resonance Standard) — without requiring raw biometric data to cross institutional boundaries. This binding positions FAVORS-BERA not merely as a security system but as a foundational infrastructure layer for resonance-informed civic governance.

5. Implementation Pathway and Experimental Design

Empirical Status: Sections 5.1–5.3 describe planned implementation phases. At time of submission (March 2026), no phase has been executed. This paper constitutes the architectural specification. Peer collaboration for empirical validation is explicitly invited via the ERES Proof-of-Work repository.

5.1 Phase 0: Simulation, Formal Specification, and Aura Calibration (Planned: 2026–2027)

Phase 0 delivers formal RSR schema specification and coherence scoring algorithm, implemented as a synthetic simulation environment. Parameters calibrated to published benchmarks [1,2] for fingerprint, voice, and retinal accuracy to establish baseline ARI coherence distributions and initial RCI band estimates.

Target simulation parameters (Python — planned implementation):

```
# Phase 0 Resonance Coherence Scoring (planned)

def resonance_coherence(ari_current, eri_delta, rci_min, rci_max,
baseline_ari):

    ari_dev = abs(ari_current - baseline_ari)

    in_band = (rci_min <= ari_current <= rci_max) and (abs(eri_delta) <=
max_drift)
```

```
    score = 1.0 if in_band else max(0.0, 1.0 - (ari_dev / (rci_max -
rci_min)))
    return score  # [0.0-1.0]; feeds RHC validation or NPR routing
```

Target synthetic baseline: ARI ~ N(0.75, 0.10); RCI initial = baseline ± 0.15; target coherence detection ≥94% at ≤5% false alert rate — consistent with comparable continuous authentication benchmarks [Cheng 2026].

Phase 0 Aura Calibration: Kirlianographic (high-voltage contact photography, Kirlian 1961 [7]) fingertip corona discharge patterns serve as the initial electromagnetic envelope proxy for ARI vector construction, providing a reproducible baseline pending GAIA EMCI full-envelope sensor development. RCI initial band estimates derived from enrollment-period variance multiplied by a security-context multiplier (Phase 0 will calibrate this multiplier empirically).

5.2 Phase 1: Prototype Sensor Integration — GAIA EMCI REEPER (Planned: 2027–2028)

Phase 1 integrates commercial biometric sensors (fingerprint scanner, retinal camera, voice capture) with the BERA RSR schema, producing live ARI/ERI/RCI measurements for a controlled cohort. Electromagnetic Aura capture transitions from Kirlianographic proxy to HRV + GSR continuous sensors [Zhao 2025].

Sensor integration governed by ERES GAIA EMCI REEPER specifications: the GAIA (Global AI Architecture) framework's Electro-Magnetic Coherence Interface (EMCI) and Resonance Energy Emission Profile Evaluation Rubric (REEPER) define hardware interface standards, signal sampling rates, noise floor tolerances, and calibration intervals required for FAVORS channel data to meet BERA ARI/ERI quality thresholds.

5.3 Validation: MIEVM AD_ON-AI and P^3 Credit-Approval Testing

Validation applies MIEVM in its extended form MIEVM AD_ON-AI: multiple independent AI analytical systems process identical RSR datasets with complementary bias profiles; convergence across ARI, ERI, and RCI instruments provides confidence in the coherence scoring model. This is consistent with published MIEVM methodology [6].

P^3 Credit-Approval testing — the primary empirical validation protocol for Phase 1:

- **P^1 Proof of Identity:** FAVORS-BERA correctly verifies enrolled individuals across all six channels within RCI tolerance, measured against ground-truth enrollment records.
- **P^2 Proof of Continuity:** Phase 0 RCI bands correctly predict observed ARI/ERI variance ranges in Phase 1 live measurements, validating RCI band calibration.
- **P^3 Proof of Access Control:** IDIPITIS pre-authentication for RT Media demonstrates zero unauthenticated session establishment, with graduated protocol responses correctly triggered by threat_context_flag values.

Full RSR datasets from Phase 1 will be made available for peer replication through the ERES Proof-of-Work repository.

6. Discussion: Implications for AI-Driven Database Design

The shift from static-record to resonance-field as the fundamental identity data structure represents a general architectural principle applicable beyond biometric security — to health monitoring, behavioral analytics, organizational performance measurement, and continuous AI-governed access systems.

The RCI band as a first-class data type is the framework's most significant database design contribution. Population-averaged thresholds — the current industry norm — produce

systematically higher false-positive rates for individuals whose biometric variation naturally exceeds average ranges. RCI makes the tolerance envelope a per-individual, continuously updated parameter, enabling higher precision with lower false-positive rates across demographically diverse populations. This directly addresses AI fairness and bias-mitigation requirements.

FAVORS-BERA motivates new database design primitives:

- **New indexing strategies:** coherence-space indexing rather than template-space indexing
- **New query types:** resonance trajectory queries, RCI band drift alerts, Perciphere boundary proximity queries
- **New consistency models:** eventual coherence rather than eventual consistency for distributed biometric systems
- **New audit patterns:** signed RHC attestation records for Explainable AI accountability in access decisions driven by coherence score rather than template match

NPR Integration: the NPR_status field in the RSR ensures that state_context negative events route to restoration pathways rather than permanent lockout. This structural embedding of restorative ethics — deviations as growth opportunities, not punishment triggers — distinguishes FAVORS-BERA from punitive security paradigms and aligns with emerging AI ethics frameworks emphasizing fairness, accountability, and transparency [8].

7. Conclusion

This paper has presented FAVORS-BERA, a unified biometric data architecture in which identity verification is reconceived as a dynamic resonance coherence problem. The framework integrates:

- Six biometric channels with forensic timestamping (FAVORS)
- Bio-energetic resonance data model with ARI, ERI, and the novel Resonance Continuity Index (BERA / RCI / RHC)
- Non-Punitive Remediation as the structural response to coherence deviation (NPR)
- Federated identity and pre-authentication protocol (IDIPITIS)
- Multi-Instrument Ensemble Validation engine (MIEVM AD_ON-AI / ARI/ERI/RCI)
- Semiosphere ontological framework for principled biometric signal-space boundary definition
- PPDM governance extensions binding RSR to UBIMIA, GCF, and NBERS frameworks

Empirical Status Recapitulation: This paper reports architectural specification, not experimental results. The contribution is a new paradigm for dynamic biometric identity verification with explicit ethical constraints and restorative remediation embedded in the data model. Phase 0 simulation and Phase 1 prototype pathways are fully specified.

This paper was produced through human-AI collaborative synthesis consistent with the ERES Institute's MIEVM methodology for multi-instrument ensemble validation with transparent attribution. Version lineage is documented in the ERES Proof-of-Work repository.

The ERES Institute for New Age Cybernetics invites peer collaboration on formal specification, simulation design, Kirlianographic calibration, and empirical validation. All foundational framework documents are available in the ERES Proof-of-Work repository at github.com/ERES-Institute-for-New-Age-Cybernetics/Proof-of-Work.

Acknowledgments

This paper was developed through iterative human-AI collaborative synthesis applying MIEVM (Multi-Instrument Ensemble Validation Method) across multiple AI systems with complementary analytical profiles (Claude, Grok, DeepSeek, USE.ai). The convergence observed across instruments on empirical honesty framing, NPR as structural design element, and RCI as primary innovation provides ensemble confidence in the Version 5 architectural specification. This attribution model is consistent with the ERES Institute's foundational principle: Don't Hurt Yourself. Don't Hurt Others. Build for Future Generations.

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